Perisha Dharel

MSE804_Blockchain Assesement

Blockchain Technology assessment -1

Document Details

Submission ID

trn:oid:::3618:112849636

Submission Date

Sep 18, 2025, 1:11 PM GMT+12

Download Date

Sep 18, 2025, 1:13 PM GMT+12

File Name

MSE804_Blockchain Assesement.docx

File Size

144.1 KB

21 Pages

5,246 Words

30,743 Characters

8% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography

Match Groups

- **19** Not Cited or Quoted 5%
  Matches with neither in-text citation nor quotation marks
- **6** Missing Quotations 1%
  Matches that are still very similar to source material
- **8** Missing Citation 2%
  Matches that have quotation marks, but no in-text citation
- **2** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

6% Internet sources
3% Publications
4% Submitted works (Student Papers)

Match Groups

- **19** Not Cited or Quoted 5%
  Matches with neither in-text citation nor quotation marks
- **6** Missing Quotations 1%
  Matches that are still very similar to source material
- **8** Missing Citation 2%
  Matches that have quotation marks, but no in-text citation
- **2** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

6% Internet sources
3% Publications
4% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	vipsolutions.in	2%
2	Internet	download.bibis.ir	<1%
3	Publication	"Fostering Machine Learning and IoT for Blockchain Technology", Springer Scienc...	<1%
4	Student papers	Universiteit van Amsterdam on 2019-07-05	<1%
5	Student papers	University of Wales Swansea on 2022-12-14	<1%
6	Student papers	BB9.1 PROD on 2025-05-09	<1%
7	Student papers	North Shore International Academy on 2025-07-12	<1%
8	Internet	www.coursehero.com	<1%
9	Student papers	University of Greenwich on 2023-11-02	<1%
10	Internet	ciptor.com	<1%

11 Internet

www.ledger.com <1%

12 Internet

www.termpaperwarehouse.com <1%

13 Publication

Alqaryouti, Omar. "Customs Trade Facilitation and Compliance for Ecommerce Us... <1%

14 Student papers

University of Glamorgan on 2018-09-30 <1%

15 Internet

dokumen.pub <1%

16 Internet

ebin.pub <1%

17 Student papers

EC-Council University on 2025-08-31 <1%

18 Student papers

University of Newcastle upon Tyne on 2023-10-17 <1%

19 Publication

Yassine Maleh, Mohammad Shojafar, Mamoun Alazab, Imed Romdhani. "Blockch... <1%

20 Publication

Zhang, Zixu. "Improving Efficiency and Security of Sharded Blockchain", Universit... <1%

21 Internet

mdpi-res.com <1%

22 Internet

peerj.com <1%

23 Internet

pure.uva.nl <1%

24 Student papers

University of Wales Institute, Cardiff on 2025-03-07 <1%

25 Publication

Francisco Liébana-Cabanillas, Francisco Muñoz-Leiva, Juan F. Prados-Castillo, Elen... **<1%**

COLLEGE OF CREATIVE INNOVATION

Programme

Master of Software Engineering – 180 credits

Blockchain and decentralised digital identity

Submitted by:

Perisha Dharel (270629406)

Course Code:

MSE804 – Blockchain and decentralised digital identity

Tutor Name:

Arshad Khan

7

Date of Submission:

21 September 2025

Yoobee Colleges of Creative Innovation

School of Technology

Auckland, New Zealand

Contents

1 Task 1 1
1.1 Task 1-A: Principles and techniques associated with blockchain tech- nology 1
1.2 Task1-B: Decentralised digital identity to meet software industry re- quirements 4
2 Task 2 8
2.1 Task2: Analyse the challenges, risks, regulations, prospects, and best practices 8
3 Task 3 12
3.1 Task3-A: Integration and importance of cryptographic techniques 12
3.2 Task3-B: Decentralised systems architectures and innovative systems and services 14
4 Conclusion 16
References 17

12

Table of Figures

Figure 1: Blockchain Structure..... 2
Figure 2: Decentralised Digital Identity..... 6
Figure 3: Risk and Mitigation 10

1 Task 1

Critically evaluate the principles and techniques associated with blockchain technology and decentralised digital identity to meet software industry requirements.

1.1 Task 1-A: Principles and techniques associated with blockchain technology

a) Assess the essence and basics of blockchain technology and their applicability in the finance industry.

Blockchain main concepts and their application in the finances The working principles of the blockchain technology are four-fold, namely, **decentralisation, transparency, immutability**, and **consensus**. Both the principles have specific connotation to the world of finance where security, trust and compliance are essential values.

- **Decentralisation** eliminates the need of having a central authority to validate transactions as it decentralises the operation of validating transactions. In the traditional financial systems, there are inefficiencies, costs and risk of systemic failure caused by the central intermediaries like clearinghouses and banks. The 2008 world financial crisis demonstrated the risks of highly consolidated control and non-transparency of centralised institutions. Blockchain approaches these risks through decentralisation and hence there lacks one point of failures. Such platforms as RippleNet follow this idea and provide almost instant cross-border payments at a much cheaper rate than the SWIFT system where many intermediaries were involved (Catalini & Gans, 2020).

- **Transparency** the confirmation of the transaction is publicly or semi-publicly visible on the block chain ledger. This property in finance ensures improved responsibilities in terms of auditing and reporting. Regulators and the auditors cannot alter transaction history as transaction history is immutable thus fraudulent activities are more difficult to conceal. As an example with transparent securities trading blockchain platforms, there would be full viewing in terms of asset ownership and trade flows, and this would decrease insider trading (Zheng et al., 2018a).

- **Immutability** is due to encryption of the difference between the blocks by hash functions and validation by consensus. Once written and verified, historical data cannot be changed. This is especially important in the area of financial audit and regulatory compliances, where records cannot be tampered with. As another example, the immutability of the record of transactions would bar manipulation that occurred with accounting scandals such as Enron where executives would use reflective record-keeping to cover losses (Narayanan et al., 2016).

- **Consensus Mechanisms** are the methods which confirm the correctness of transaction in the distributed network. Power, security and authority is not controlling by another person or entity, unlike in the traditional financial system where trust is based on centralised clearing houses. Other types, including Proof of Work (PoW) and Proof of Stake (PoS) are not without several trade-offs with respect to security and performance. Consensus mechanisms also enable handling of assets without third-party involvement, which may be applied to finance through faster and more energy-efficient execution of financial smart contracts, such as in the case of the Ethereum transition to PoS (Androulaki et al., 2018).

Overall, blockchain principles target crucial financial industry issues, i.e., fraud deterrence, compliance, cutting the costs of cross-border settlements, and trust which makes it a game-changing technology in the industry.

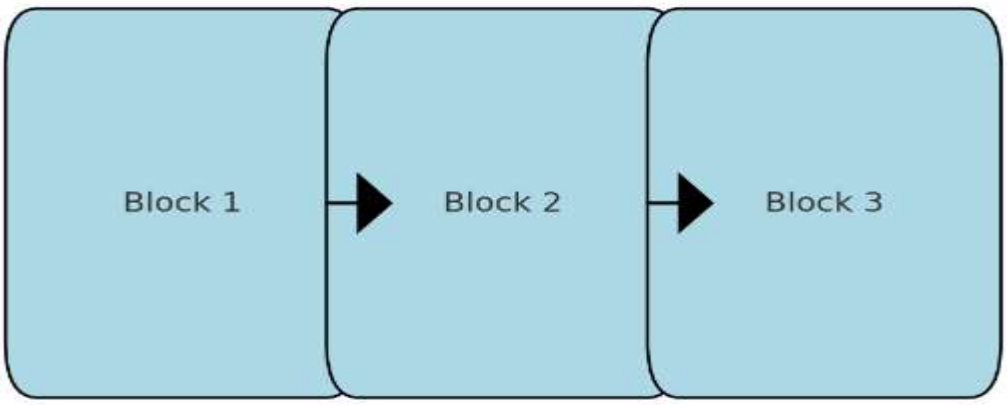

Figure 1: Blockchain Structure

Cryptographically been linked chain of blocks. Transactions are stored in each block and once added, prior blocks cannot be modified in order to provide an immutable nature.

b) Explain and justify all the factors which make blockchain an immutable ledger.

The immutability of blockchain is not a feature but the side-effect of three mechanisms which interact in a synergistic neural fashion **cryptographic hashing, consensus protocols**, and **replication of distributed ledgers**.

- **Cryptographic Hashing:** Because it cryptographically hashes each block of input data, the block data can only be irreversibly converted into a fixed-length hash (typically SHA-256). The next block has a header with a hash of the previous block to create a chain of hash. When one transaction is changed by an attacker, the hash values become cascaded, so thereafter the blocks are invalid. This allows computationally viable tampering to be detected (Narayanan et al., 2016).

- **Consensus Protocols:** Consensus protocols are the ones that enforce high levels of immutability since new blocks cannot be added to the blockchain until there is an agreement within the network. To give one example, in Proof of Work systems one can only alter history by re-mining all subsequent blocks and overcoming the honest network in terms of the sheer amount of hash power available to it–which is impractically difficult unless one controls the majority of hash power in the network in the first place (Conti et al., 2018). In Proof of Stake attackers would have to possess majority of tokens and manipulation would be extremely expensive.

- **Replication of distributed ledger:** It also augments immutability Blockchain information is sent to numerous nodes throughout the world. To make tampering effective, an attacker would need to manipulate most of these distributed copies made all at once and this is unrealistic (Zheng et al., 2018a).

All these make blockchain an append-only database; once a record has been written it cannot be destroyed or changed. It is through this immutability that the financial auditing-, cross-border settlements-, and regulatory reporting can be trusted, through blockchain.

c) Explain and justify with an example how 'Speed' is a weak point of Public Blockchain (w.r.t. TPS rate) but a strong point of Private Blockchain?

One of the key obstacles faced in the use of blockchain in financial activities is speed of trans- actions. There is a large difference between the throughput in public and private

blockchains.

Such approval mechanisms on public blockchains as PoW with participation globally lead to 7 to 30 transactions per second (TPS). They are also inappropriate to use in places where daily transactions are highly active like retail payment or trading securities where transactions currently exceed **65,000 TPS** by the Visa network (Narayanan et al., 2016). This time lag between the initiation and the ultimate confirmation of a transaction (usually few minutes) makes it cumbersome to provide real-time transactions in financial services.

Privately held blockchains on the other hand are optimised in terms of performance. Other systems, such as **Hyperledger Fabric**, have permissioned access and are based on a consensus system, such as Practical Byzantine Fault Tolerance (PBFT). These systems do not require so much energy as the mining process and the number of participants that are involved in the validation process is far fewer, allowing up to thousands of TPS (Androulaki et al., 2018). An example of such innovations is the **JPMorgan's Quorum blockchain** , which allows settlements between banks to be executed at the same speed as in a conventional clearinghouse with added transparency and security.

In short, speed is a significant drawback of blockchains and a positive aspect of privately operated or consortium blockchains, making the latter a better choice to enterprises when it comes to financial applications (Zheng et al., 2018a).

1.2 Task1-B: Decentralised digital identity to meet software industry requirements

a) What is a 51% attack in a blockchain environment and how it can be mitigated?

5 A 51% attack is a type of attack wherein the attacker gets possession and control over more than half of the mining or validation power in a blockchain. By obtaining a majority, fraudsters can undo transactions, repeatedly spend tokens and stop the confirmation of new blocks (Conti et al., 2018).

A 51% attack is devastating in the financial terms.
As an example, In 2019 the Ethereum Classic blockchain was attacked in a 51% attack where over USD $1.1 million (Zhang et al., 2019) was double-spent. Hackers acquired more than half

of the mining capabilities of the network, enabling them to reverse validated transactions and use the same coins twice. This made smaller Proof of Work blockchain susceptible to 51percent attacks, which result in financial loss and loss of confidence in the network.

The mitigation methods may be as follows:

- **Proof of Stake (PoS):** In PoS, the rationality of attack would entail the attackers having majority of the tokens and this will not be economically feasible.
- **Hybrid consensus algorithms:** The combination of PoW and PoS leads to better resilience as attacking them may be diversified.
- **Level of network decentralisation:** The extra level of nodes and geographical dispersions reduces the possibility of collusion.
- **Monitoring tools:** Blockchain networks are capable of implementing real-time anomaly detectors to identify the unusual behaviour of mining action (Yaga et al., 2019).

These strategies will help prevent the risk of 51% attacks, thus providing security to financial transactions.

b) Define decentralised digital identity and discuss why it is important to solve identity management issues in the finance sector.

Decentralised digital identity (DID) are identity systems, whereby the management of identifiers and credentials lies in the hands of individuals, and the identity is authenticated by blockchain trusted nodes. Whereas centralised identity systems are liable to data breaches by relying on single institutions, DID does not rely on any specific institutions and as such is not susceptible to data breaches in the same way (W3C, 2022).

In the financial sector, DID lowers issues in identity management that are persistent:

- **KYC/AML compliance:** Banks and other financial institutions knowingly pay billions of dollars every year on Know Your Customer (KYC) audits. This pilot will help to realize verifiable credentials that can be reused across financial institutions, which will seriously cut down the compliance expenses (Preukschat & Reed, 2021).

- **Fraud prevention:** DID has that advantage in as much as cryptographic proofs being resistant to forgery lower the chances of identity theft or impersonation.
- **Cross-border interoperability:** DID systems operate across the border and so, one customer can use financial services in other countries without going through the process of identity verification once again.

Therefore, DID is lower in cost, and highly secure to the financial institutions.

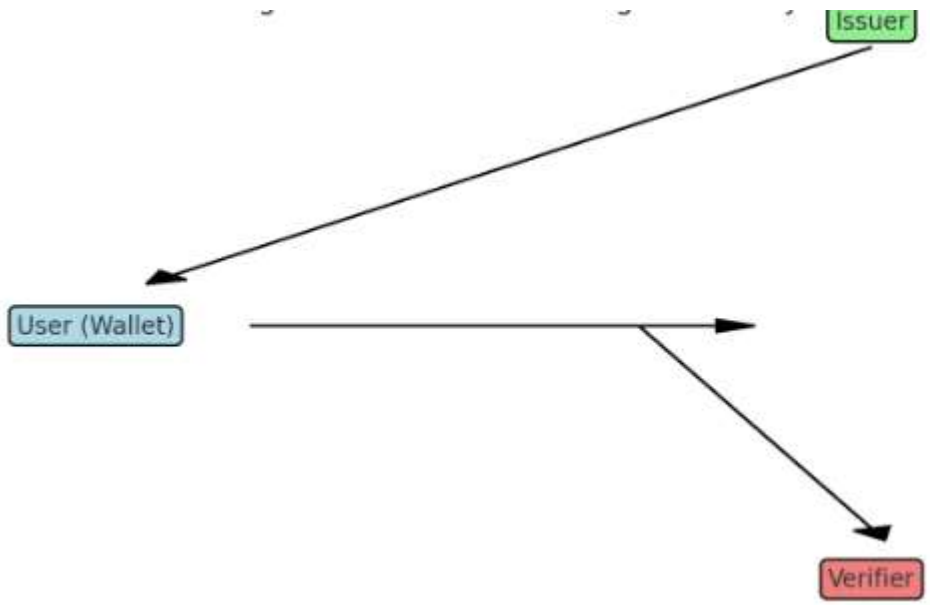

Figure 2: Decentralised Digital Identity

Above figure shows the DID model:

- Verifiable credentials are issued by issuer.
- They are kept in a digital wallet by user.
- Which authenticates verification checks in a blockchain that has no coordinator.

c) Consider the possible effect of digital identity on the improvement of security and privacy in banking system.

Adoption of DID can provide important security and privacy to the banking system.

First, DID minimises the use of centralised identity databases which are common target of cyberattacks. One can mention the Equifax data breach (2017) when personal information of 147 million

people was exposed mainly due to the vulnerability of storing information in centralised locations (Allen et al., 2020). DID minimises this by distributing storage of credentials in which the persons have control over their identities by using secure digital wallets.

Second, DID increases privacy by use of selective disclosure. To illustrate, a customer of the bank could confirm his/her age exceeds 18 years without indicating his/her date of birth, and therefore meet privacy-by-design principles in light of GDPR. This will impede data over-collection and reduce exposure to risk (European Union, 2023).

Lastly, DID improves trust in users. Restoring the ownership of digital identify back to customers, banks send the signal that they are willing to protect the autonomy of users. It is especially essential in the context of an era when the trust of the consumers in financial institutions is worn down by the repeated scandals involving data (Preukschat & Reed, 2021).

In summary, it is possible to say that DID provides not only better security and privacy but also competitive advantage to the banks due to consumer trust generated by DID.

2 Task 2

Evaluate the obstacles, hazards, policies, opportunities, and good practices that revolve around the concept of decentralised digital identity and blockchain applications in software design solutions.

2.1 Task2: Analyse the challenges, risks, regulations, prospects, and best practices

a) Trace and assess the main issues regarding the adoption of decentralised digital identity in software design.

The application of the decentralised digital identity (DID) to software systems poses important technical, organisational and regulatory issues.

- **Cross standard inter-operability:** There is lack of inter-operability among standards in DID solutions. Protocols Competing products include the W3C, Sovrin, Hyperledger exposed to the Indy (project), and the Microsoft ION that operate on different technical frameworks, where cross-platform adoption is hard. In the absence of interoperability, users can experience what is known in the sector as identity silos, i.e. the credentials they could use in one ecosystem, but not in the other (Wang & De Filippi, 2020).

- **Scalability and Performances:** Public blockchains appear to have limitations to throughput, excessive latency, and transaction-costs (Zheng et al., 2018a). The maintenance of mass identity registries that deal with millions of users raises the issue of scaleability. As another example, the gas fees on the Ethereum network at times of congestion render the anchoring of identities on-chain financially infeasible.

- **Regulatory ambiguity:** Governments are finding it hard to define how decentralised identities should be classified within existing legislations. To illustrate, the right to be forgotten issue in GDPR does not sit well with blockchain immutability because it is not easy to delete an individual's personal data (European Union, 2023). Similarly, to implement financial regulations such as AML/KYC, there is the need to understand the nature of DID frameworks against auditing requirements.

- **User acquisition and availability:** The process of managing the private keys and digital

wallets is difficult to non-technical users. The misunderstanding with a lost or stolen private key may lead to irreparable loss of identity credentials (Allen et al., 2020). This constitutes a hindrance to adoption in general populations, particularly in the developing world.

- **Governance issues:** There are still questions on how credentials are issued, revoked or even updated when there is one system. In the absence of apparent governance, the trust anchors may be corrupted with this providing a reductions in institutional acceptance (Preukschat & Reed, 2021).

These issues point to the idea that DID is a potentially transformative technology that needs to be standardised at the technological level, be more defined regulatorily, and adjusted to the needs of users before its widespread use in the financial sector and beyond.

b) Analyze the possible risks and security issues with the implementation of the blockchain into the software design solutions. In addition, comment on plans and optimal practices to reduce risks associated with blockchains.

The incorporation of blockchain into software design involves technical, operational, and legal risks, so they need to be addressed efficiently.

- **Privacy risks:** The elements that are published in a blockchain consist of time of transaction and wallet addresses. Although the personal identifiers may be encrypted, metadata analysis can be used to deanonymise users (Zhang et al., 2019).

  Mitigation: Putting sensitive data on off-chain storage with blockchain hashes on so-called anchors. An alternative to validation of information that does not require any disclosure of sensitive information is zero-knowledge proofs (ZKPs).

- **Smart contract flaws:** Flawed Smart Contracts are subject to exploits. In 2016, the attacks on the DAO entity took advantage of the vulnerability due to a recursive call and went on to steal USD $60 million (Narayanan et al., 2016).

  Mitigation: Carry out formal inspection and peer audits and automated vulnerability checks using security solutions like MythX.

- **Key management risks:** Data loss that cannot be reversed is caused by loss of keys and the attacker is provided with the complete control by compromised keys.

  Mitigation: Distributors Use multi-signature wallets, custodial recovery and hardware security modules (HSMs).

- **Regulatory and compliance risks:** Blockchain is an inter-jurisdictional space and regulations differ. An example is where cryptocurrencies are prohibited in one country and permitted legally in another country.

  Mitigation: Embrace financial regulators turning to the use of regulatory sandboxes in which experimentation can be enjoyed within a specific environment.

- **Operational risks:** Executing blockchain by connecting with legacy systems is technically demanding, and might entail a process of redesign and retraining of employees (Yaga et al., 2019).

  Mitigation: Hybrid architectures that integrate blockchain with the existing infrastructure should be used to help perform a slow migration.

The adoption of the best practices will allow organisations to potentially eliminate risks to the largest extent possible and gain the most of the benefits that blockchain has to offer.

Risk	Mitigation
Privacy Risks	Off-chain storage, ZKPs
Smart Contract Flaws	Formal audits, tools
Key Management	Multi-sig, HSM
Regulation	Sandboxes, compliance
Operations	Hybrid integration

Figure 3: Risk and Mitigation

Can be found above table among most common risks (privacy, smart contract defects, loss of keys, regulatory, operational) and mitigation mechanisms (ZKPs, audits, multi-sig wallets, sandboxes, hybrid integration).

c) Name and explain best practices in effective and successful implementation of decentralised digital identity in software design. Besides these, discuss instances of organisations or projects that have succeeded in implementing best practices of digital identity.

The implementation of the DID should go in tandem with the technical standards, regulatory and user experience practices.

- **Use of open standards:** W3C DID Core and the use of Verifiable Credential are open standards that make them recognised across the globe thereby ensuring interoperability between ecosystems (W3C, 2022).

- **Layered architecture :** Storing sensitive characteristics of identity in off-chain storage, and verifiable proofs in on-chain storage. This also minimizes when personal information is disclosed and that privacy rules are adhered to (Zhang et al., 2019).

- **User-centric design:** The DID wallets are to facilitate recovery mechanisms, including the social key recovery in which peer trust helps to recover the credentials that were lost (Allen et al., 2020). If the interface is made simpler to the non-technical users, then there is likelihood of adoption.

- **Industry cooperation:** Real-life practical applications with verifiable credentials and DIDs are already proven by organisations like Sovrin Foundation and banks such as ING, which are currently using DIDs to onboard their customers and stay compliant with anti-money laundering (AML) and know your customer (KYC) requirements(Preukschat & Reed, 2021).

- **Regulatory push:** Digital identity wallet is presented formally through the European EIDAS 2.0 regulation, and it can result in the global leadership (European Union, 2023). Conformity with such frameworks raises the levels of legitimacy and use. Organisations are able to use these practices to implement DID systems which are scalable, compliant, user-friendly, and secure and are therefore implementable in real life.

3 Task 3

Innovate systems and services with blockchain technology, cryptography, systems architecture that is decentralised, and information systems.

3.1 Task3-A: The incorporation and significance of cryptographic methods

1

a) Give an explanation of how cryptography is used to provide security of the integrated systems. In addition, discuss cryptographic techniques used to protect data integrity, confidentiality, and authentication such as hash functions, digital signature, merkle tree, Elliptic Curve Cryptography (ECC) and byzantine fault tolerance.

Cryptography provides security by assuring confidentiality, integrity, authentication, and fault-resilience as the framework of blockchain.

- **Hash functions (e.g. SHA-256):** In order to guarantee data integrity blockchain transactions are transformed into tamper-evident cryptographic hash functions. A transaction is carried out to create a hash value. When any aspect of the transaction is changed even slightly, the hash will change and immediately alert the system that the information has been manipulated or corrupted. This helps the network to detect discrepancies and offer reliability and trustworthiness of the blockchain registry (Narayanan et al., 2016).

- **Digital signatures:** The transfer of assets through blockchain is insured using the means of private keys, thus it is only the owner of the assets to make the transfer. The mechanism helps to prevent access controls, identity theft and phishing, thus ensuring integrity, authenticity and safety of financial systems. The reason why the private keys are vital in a trusted online transaction is that they verify the actual owner of the key and no sensitive information is given out.

- **Merkle trees:** Merkle trees facilitate the sound verification of large amount of data through hierarchical hashing of data. Particularly well with lightweight clients, which do not need to download the whole blockchain, they are able to check the validity of transactions. The Merkle trees use a hash path verification method to reduce dramatically the computational and storage costs by not verifying the entire dataset but only the relevant hash paths (Zheng et al., 2018a).

20

- **Elliptic Curve Cryptography (ECC):** ECC is a form of public-key cryptography, which is based on the mathematical characteristics of an elliptic curve to lock data. It is very efficient in bandwidth, storage and calculation, and provides powerful encryption on very small key size than the traditional encryption algorithms, including RSA. Digital signatures are created in ECC in blockchain networks and the transactions are authentic, secure, and tamperproof (Zhang et al., 2019).

- **Byzantine Fault Tolerance (BFT):** BFT is a distributed system consensus mechanism that guarantees that the system works correctly even when some of the nodes are malicious or random. It is resilient to one-third of participants being compromised, and, nonetheless, allows the network to reach consensus on what the right state is. BFT is commonly used within permissioned blockchain networks to guarantee that the records of transactions are consistent, reliable, and resistant to attacks (Androulaki et al., 2018).

In a combination, these cryptographic methods guarantee trustless security of distributed systems, which is essential to such sectors as banking and healthcare.

b) Compare and contrast different consensus algorithms used in blockchain networks (e.g., Proof of Work, Proof of Stake).

Consensus protocols are used to ensure congruence amongst states of ledger across decentralised nodes.

- **Proof Of Work (PoW):** This is highly secure, and it makes blocks to be created with effort applied through computations. But then it is power-hungry and glacial (7 TPS in Bitcoin). Its ability to defy attacks surpasses none yet there are great environmental and cost trade-offs (Narayanan et al., 2016).

- **Proof of Stake (PoS):** The validators are chosen based on their possession of tokens, thus it is energy efficient and faster (up to 1000+ TPS). Critics cite that PoS is susceptible to centralisation as more money would grant the participating node additional power (Zhang et al., 2019). In 2022 Ethereum transferred to PoS which demonstrates its affordability to larger-scale applications.

- **Delegated Proof of Stake (DPoS):** Delegates who are elected by the society verify the transactions. It is very through-putding and less decentralizing. One of the examples is EOS, which can manage thousands of TPS (Wang & De Filippi, 2020).

Algorithm	Security	Performance	Trade-offs	Example
PoW	High	Low (7 TPS)	Energy-intensive	Bitcoin
PoS	Moderate	High (1000+ TPS)	Centralisation risk	Ethereum 2.0
DPoS	Moderate	Very High	Less decentralised	EOS

The comparison of these mechanisms shows that there is really no perfect consensus, as all of them have some sacrifices concerning primitivisation of security, efficiency, and decentralisation.

c) Explain the importance of consensus in maintaining the integrity of the blockchain.

The importance of consensus to the integrity of blockchains is that it is what guarantees that all the actors in the network share a common, agreed upon version of the ledger. This agreement will help avoid any form of fraud, like the one of spending money twice and will not allow the chain to fork into different versions or sub-versions (Nakamoto, 2008; Zheng et al., 2018b).

The use of consensus in financial and digital asset solutions is that once a transaction has been validated, it cannot be changed or reversed and is therefore accountable and dependable (Gervais et al., 2016). Further, consensus protocols (such as Proof of Work (PoW) and Proof of Stake (PoS)) remove the necessity of centralised intermediaries, enabling the existence of decentralised networks that could operate securely with the maintenance of transparency, fairness and trust among network participants (Cachin & Vukolić, 2017).

3.2 Task3-B: Decentralised systems architectures and innovative systems and services

a) Provide and outline new applications of blockchain technology that have been successfully applied to systems and services. In addition, speak about the influence of blockchain on the improvement of security, transparency, and efficiency.

Blockchain has enabled the technology in different sectors:

- **Trade Finance:** The global shipping industry uses TradeLens system developed by IBM to digitise shipping papers on blockchain and improve transparency, minimise frauds and streamline customs procedures (Hackius & Petersen, 2017).

- **Healthcare:** Estonia uses blockchain to store the med files of its citizens which guarantees confidentiality but provides transparency to data entrants. This enhances security and trust of healthcare management information (World Bank, 2019).

- **Banking:** JPMorgans Quorum blockchain is facilitating faster Interbank transfers and can minimize reconciliation requirements showing effectiveness and improvements in reliability of transactions in the capital markets.

- **Digital Identity:** The Sovrin Network offers decentralised identity solutions and enable individuals to attest attributes (e.g. age, citizenship) in a privacy-preserving way without disclosing personal information which can add privacy, security and control (Preukschat & Reed, 2021).

Implications in Other Sectors: The cases reveal that blockchain boosts security due to its immutability and cryptography, transparency since the participants can audibly and tamper-proofly analyze the records, and efficiency by reducing the number of intermedi- aries, automation of processes and operational costs. All these demonstrations prove that blockchain is a revolutionary technology in various sectors.

b) Explain the effects of the decentralised systems architectures on the design and development of innovative information systems.

The decentral architectures paradigm extreme re-architectures the software design by incorporating decentralised control because it does not use centralised servers. Such a move influences design in the following ways:

- **Resilience:** There are no points of failure with decentralisation and hence make systems more resilient and resistant to any form of outage or attacks. Less prone to server outages than centralised cloud-storage, such as AWS, decentralised cloud-storage networks including Filecoin store the data in more than one node.

- **Transparency:** Blockchain registries enable various parties to communicate and confirm transactions without using an intermediary. This mistrustful atmosphere has allowed everyone to verify the accuracy of transactions on their own, which promotes responsibility.

- **Security:** When the network is decentralised, the attackers will not be able to simply bring the network down by targeting one node. To be successful, they have to have

control over a large part of the network, which is computationally expensive and practically hard, thus improving the overall security.

By extension, decentralised models will also compel developers to focus on fault-resilience, trustless cooperation, and decentralised governance (Narayanan et al., 2016).

c) Explore the trade-offs and benefits associated with decentralised design principles.

Decentralised design principles share the control, data and decision making among the different nodes which have their share of benefits. Advantages comprise increased transparency, where all transactions can be seen and tracked; high levels of security due to their cryptographically validated immutability and the reduced need to depend on a third party; peer-to-peer interactions such as in decentralised finance (DeFi); and greater control over digital assets and data on the part of the individuals(Preukschat & Reed, 2021).

These gains however, have trade-offs. Decentralised networks tend to be slower and are also less scalable, as shown when Ethereum became congested as NFTs started to find traction (Zheng et al., 2018a). High-energy consumption consensus, complicated governance, and unclarity of regulation are also an issue. Multiple node decision-making takes longer in some cases and enforcement of accountability can be difficult in contrast to centralised systems. Hybrid systems, e.g. permissioned blockchains, try to trade off these aspects as they combine decentralisation with efficiency and government control, are less vulnerable and easier to monitor, but have a faster transaction speed and greater compliance. Overall, decentralised design principles have the potential to provide resilient, trustless solutions, yet it has to be supported by planning of the architecture and governance to maximise the benefits and counterbalance the limitations.

4 Conclusion

Blockchain and decentralised design principles have the potential to be transformational across industries since it facilitates secure, transparent and efficient systems. Consensus mechanisms (Gervais et al., 2016; Nakamoto, 2008), allow avoiding outside parties, to ensure the integrity of the ledger, to prevent fraud, and to engender trust in the decentralised network (Cachin & Vukolić, 2017). Expanding adoption of blockchain technology has taken place in various fields

such as trade finance and healthcare, banking, and digital identity, where innovative applications prove to improve transparency, build greater security, and streamline operational processes (Hackius & Petersen, 2017; Preukschat & Reed, 2021; World Bank, 2019). Meanwhile, decentralising means introducing certain trade-offs, such as issues of scalability, excess cost of computation, causing complex governance, and regulatory risks (Zheng et al., 2018a). To help overcome these trade-offs, new methods (or hybrids) are emerging e.g., permissioned blockchains, which still allow the core benefits of decentralisation. Overall, the implementation of blockchains is labor intensive and rewarding to user oriented, hard, and non-trusting systems.

References

Allen, C., Preukschat, A., & Cameron, K. (2020). The path to self-sovereign identity. *Journal of Identity Management*, *12*(1), 5–22.

Androulaki, E., Barger, A., Bortnikov, V., Cachin, C., Christidis, K., De Caro, A., Enyeart, D., Ferris, C., Laventman, G., Manevich, Y., Muralidharan, S., Murthy, C., Nguyen, B., Sethi, M., Singh, G., Smith, K., Sorniotti, A., Stathakopoulou, C., Vukolić, M., . . . Zou, J. (2018). Hyperledger fabric: A distributed operating system for permissioned blockchains. *Proceedings of the Thirteenth EuroSys Conference*, 1–15. https://doi.org/10.1145/3190508.3190538

Cachin, C., & Vukolić, M. (2017). Blockchain consensus protocols in the wild. *arXiv preprint arXiv:1707.01873*. https://arxiv.org/abs/1707.01873

Catalini, C., & Gans, J. S. (2020). Some simple economics of the blockchain. *Communications of the ACM*, *63*(7), 80–90. https://doi.org/10.1145/3359552

Conti, M., Kumar, S., Lal, C., & Ruj, S. (2018). A survey on security and privacy issues of bitcoin. *IEEE Communications Surveys & Tutorials*, *20*(4), 3416–3452.

European Union. (2023). Decentralised digital identity: Eu strategy and guidelines [Accessed: 2025-08-24].

Gervais, A., Karame, G. O., Wüst, K., Glykantzis, V., Ritzdorf, H., & Capkun, S. (2016). On the security and performance of proof of work blockchains. *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*, 3–16. https://doi.org/10.1145/2976749.2978341

Hackius, N., & Petersen, M. (2017). Blockchain in logistics and supply chain: Trick or treat? In

W. Kersten, T. Blecker, & C. M. Ringle (Eds.), *Digitalization in supply chain management and logistics: Smart and digital solutions for an industry 4.0 environment* (pp. 3–18). epubli. https://doi.org/10.15480/882.1444

Nakamoto, S. (2008). Bitcoin: A peer-to-peer electronic cash system.

Narayanan, A., Bonneau, J., Felten, E., Miller, A., & Goldfeder, S. (2016). *Bitcoin and cryptocurrency technologies: A comprehensive introduction*. Princeton University Press.

Preukschat, A., & Reed, D. (2021). *Self-sovereign identity: Decentralized digital identity and verifiable credentials*. O'Reilly Media.

W3C. (2022). Decentralized identifiers (dids) v1.0 [Accessed: 2025-08-24].

Wang, F., & De Filippi, P. (2020). Self-sovereign identity in a globalized world: Credentials-based identity systems as a driver for economic inclusion. *Frontiers in Blockchain*, *2*, 28. https://doi.org/10.3389/fbloc.2019.00028

World Bank. (2019). Digital-in-health: Unlocking the value for everyone.

Yaga, D., Mell, P., Roby, N., & Scarfone, K. (2019). *Blockchain technology overview* (NISTIR 8202). National Institute of Standards and Technology (NIST). https://doi.org/10.6028/NIST.IR.8202

Zhang, R., Xue, R., & Liu, L. (2019). Blockchain-based decentralized identity management. *IEEE Access*, *7*, 162139–162152.

Zheng, Z., Xie, S., Dai, H.-N., Chen, X., & Wang, H. (2018a). An overview of blockchain stechnology: Architecture, consensus, and future trends. *IEEE Access*, *6*, 55436–55448. https://doi.org/10.1109/ACCESS.2018.2872672

Zheng, Z., Xie, S., Dai, H.-N., Chen, X., & Wang, H. (2018b). Blockchain challenges and opportunities: A survey. *International Journal of Web and Grid Services*, *14*(4), 352–375. https://doi.org/10.1504/IJWGS.2018.095647